pfi

Python for Fixed Income

COURSE 5B · TIME SERIES MODELS

Case Study: AR and ARIMA on the Curve

WEEK 1 · DAY 5

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Today: two parts

Hands-on, on the 10 Yr

1. Fit an AR(1) by hand and with statsmodels
2. Choose an order: the AIC and BIC grid
3. Read every setting of ARIMA
4. Check the residuals
5. Score walk-forward

Case study, on three series

The 10 Yr, the 2s10s slope, and the invented IG spread, on day 2's five folds, each model against the last value carried forward. Development years only: 2 Jan 2015 to 30 Dec 2022, 2,001 dates.

The question: does a model of the change's own past beat carrying the last value forward?

Names for a model's output

Today is the first day with models. A model's output is a fitted change or a one-step value for a date inside the file. The baseline:

Allowed names

fitted change in bp

the model's one-step value

the last value carried forward

Never called

a forecast or a prediction

a projection or an outlook

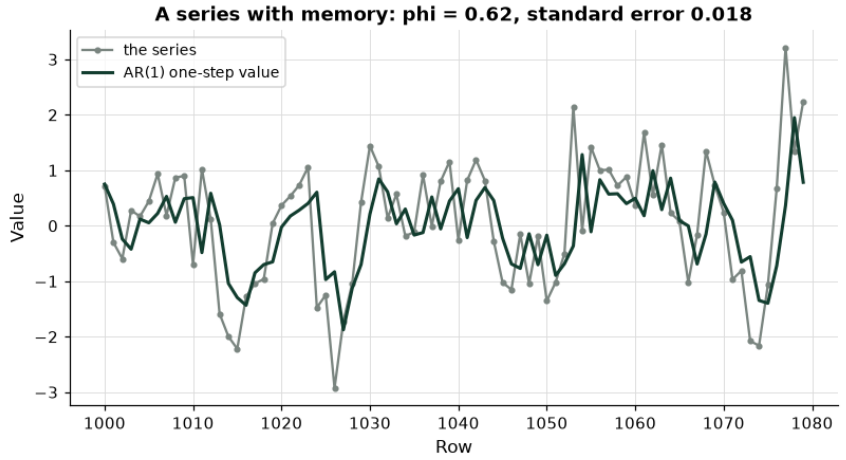
a signal, a call, a view on rates

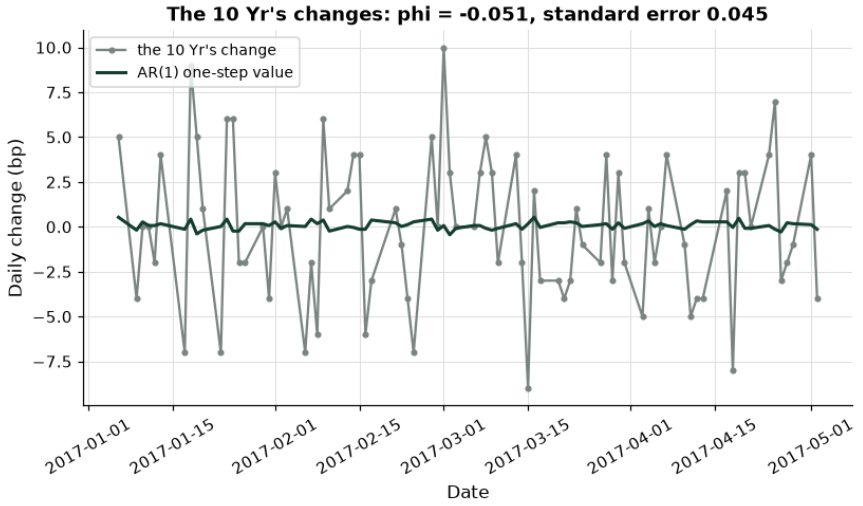
Every score is on dates already in the file, from a model fitted on earlier dates.

How AR, MA, and ARIMA work

AR(p)	A value is a constant plus p weights times the p values before it, plus a shock: it remembers values
MA(q)	A value is a constant plus a shock plus q weights times the q shocks before it: it remembers surprises
ARIMA(p, d, q)	Difference d times (day 3), then fit an AR(p) and an MA(q) to what is left
The fit	Maximum likelihood: coefficients until the implied shocks look like independent noise
What comes out	Coefficients, shock variance, AIC and BIC, residuals, and the one-step value for any row

A series with memory, and the 10 Yr





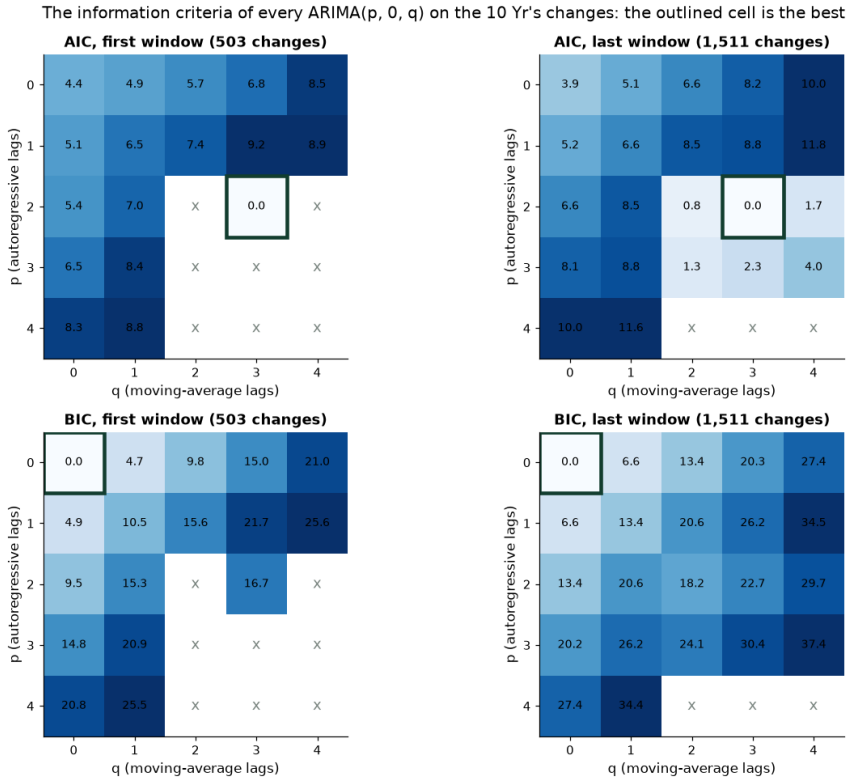
Method	phi	constant
least squares by hand	0.622370	-0.052164
AutoReg	0.622370	-0.052164
ARIMA (maximum likelihood)	0.622053	-0.052109

Simulated: phi 0.6224, standard error 0.0175. The 10 Yr: phi -0.0514, standard error 0.0446, t -1.15. Fold 1 MAE: AR(1) 2.7574 bp, last value 2.7619 bp.

How information criteria and the grid work

The problem	More lags always fit better, so fit alone cannot choose the order
AIC	2k less twice the log-likelihood, k the fitted parameters
BIC	k ln(n) less twice the log-likelihood: it charges more, so it prefers shorter models
The grid	Fit every ARIMA(p, 0, q) up to a size, read the criterion, take the lowest
What comes out	A table of criteria, and the fits that did not converge, kept and counted

The grid on the 10 Yr



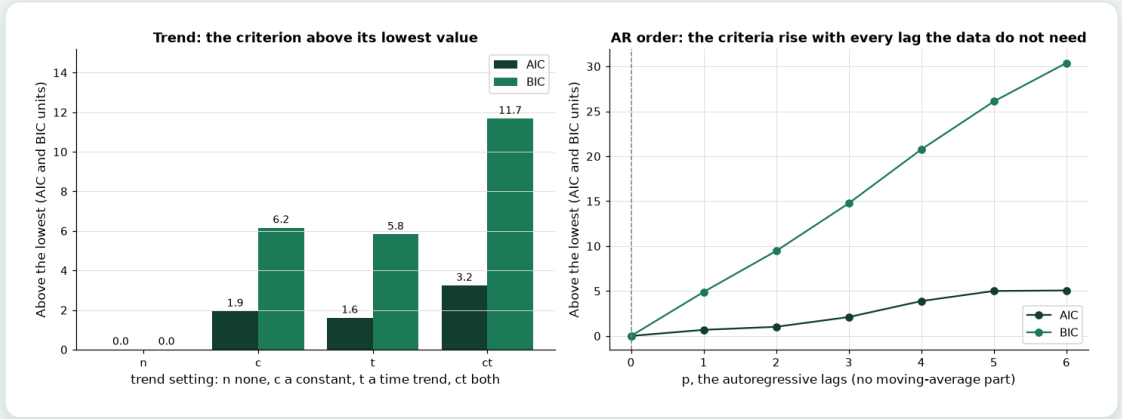
Window	BIC	AIC	Failed
first window	(0, 0)	(2, 3)	8
last window	(0, 0)	(2, 3)	3

BIC picks no lags: a constant, day 2's drift. AIC picks a longer order, and AIC's picks are the ones that fail to converge later. Orders that failed to converge are x marks, counted.

The size of the grid is a choice: a bigger grid means more fits that fail.

Every setting of ARIMA

Setting	Default	Tune first
order	(0, 0, 0)	yes
trend	None	yes
seasonal_order	(0, 0, 0, 0)	no
enforce_stationarity	True	yes
enforce_invertibility	True	no
concentrate_scale	False	no



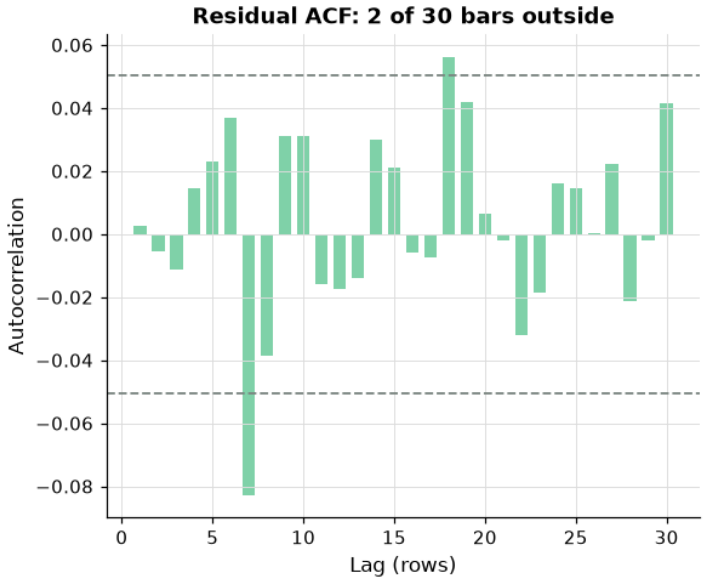
Random walk, enforce_stationarity on: phi 0.9888; off: 0.9908. d = 1 on the levels is the same model: phi -0.0513 either way.

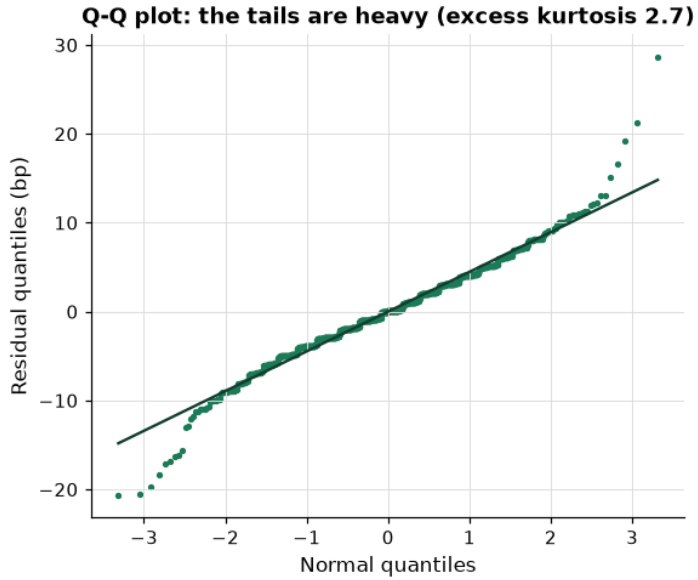
Pass the trend on every fit: trend=None means a constant only when d is 0. d = 1 on the level is an AR(1) on the change.

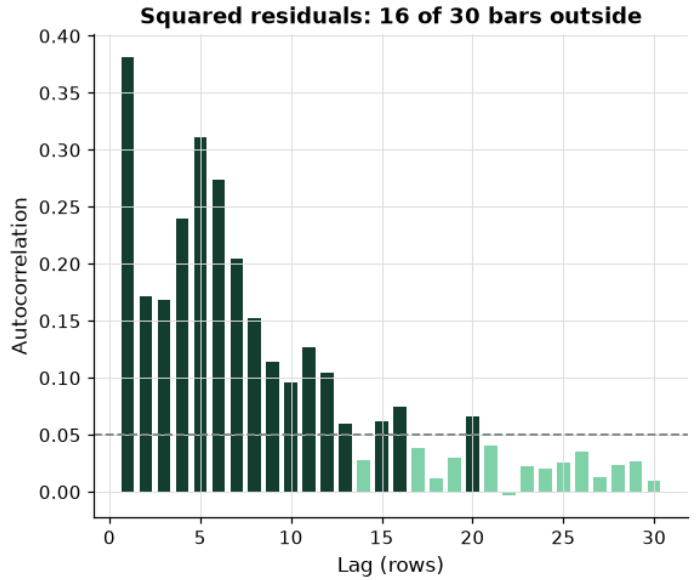
How residual diagnostics work

The idea	A model that has caught the structure leaves residuals that look like independent noise
Residual ACF	No bar should be outside the band
Ljung-Box	The joint test, with model_df = p + q: fitted coefficients use up degrees of freedom
Q-Q plot	Residual quantiles against a normal's: tails that bend away mean heavy shocks
Squared residuals	Ljung-Box on them asks whether the size of the shocks has memory

What the residuals say







ARIMA(1, 0, 1) on the last window: phi 0.5009, theta -0.5258, nearly cancelling. Residual ACF: 2 of 30 bars outside. Ljung-Box (model_df 2) p 0.0146 at 10 lags. Excess kurtosis 2.65; 0.9% of residuals beyond 3 standard deviations (normal: 0.3%). Squared residuals: p 4.5e-163.

An ARIMA assumes shocks of a constant size: the memory is in the size, and day 7 models that.

How walk-forward ARIMA works

Each fold	Fit once, on the training rows only
The test rows	Run the fitted model over them with .append(..., refit=False): coefficients do not change
The value	.predict gives the model's one-step value for each row from the rows before it
The next fold	Fitted again on a longer window
The proof	From day 2: the values must not move when the rows from the middle of the fold on are removed

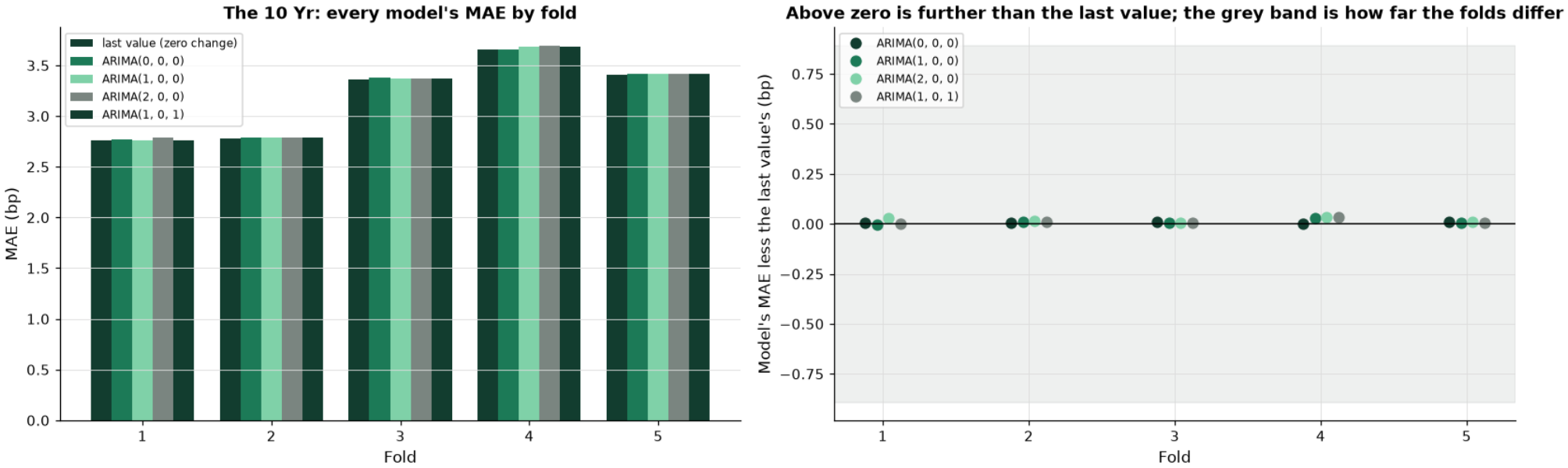
Four models against the last value

Model	Mean MAE (bp)	Gap (bp)	MASE	Folds below the last value
last value	3.1937	0.0000	1.0000	0
ARIMA(0, 0, 0)	3.2012	0.0076	1.0024	0
ARIMA(1, 0, 0)	3.2026	0.0089	1.0028	1
ARIMA(2, 0, 0)	3.2126	0.0189	1.0059	0
ARIMA(1, 0, 1)	3.2048	0.0111	1.0035	0

The 10 Yr, in bp, on day 2's five folds. The constant model is day 2's drift. Every MASE is above 1, and more parameters make it a little worse, not better.

The last value carried forward is closest. The marked cell ranks five rules on these dates, never a security.

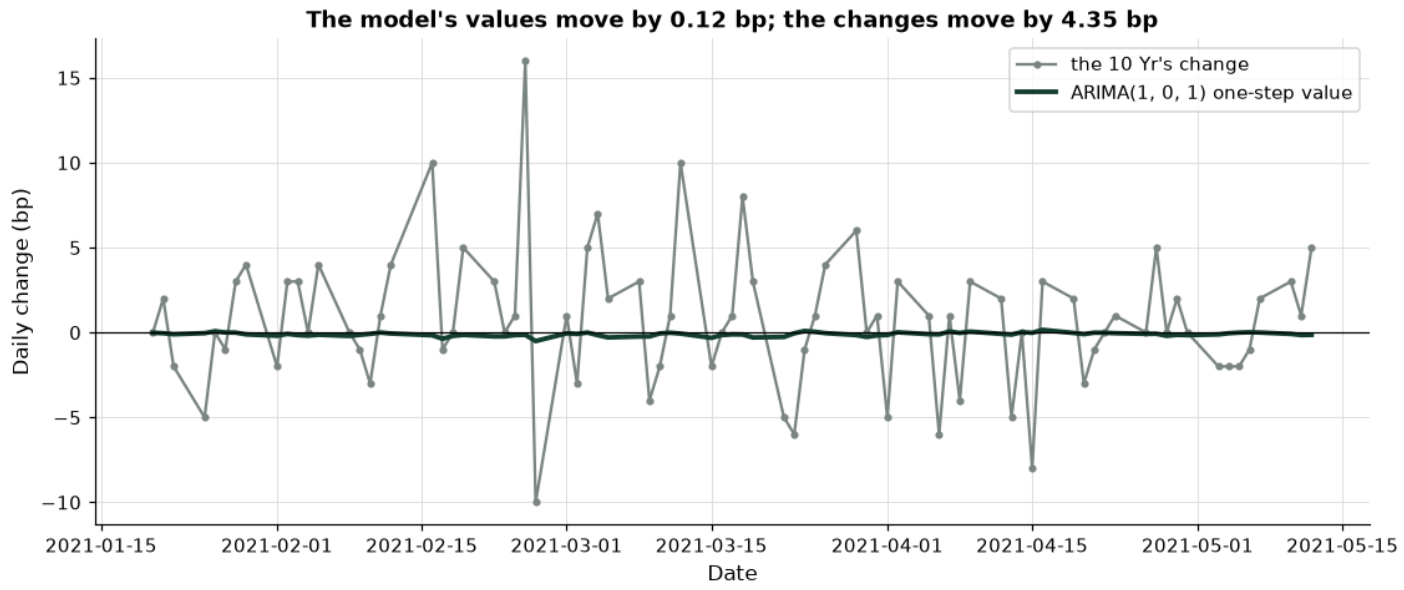
Fold by fold



Left: the bars of a fold are almost the same height for every model. Right: each dot is a model's MAE less the last value's on one fold, inside the grey band of how far the folds differ.

A model whose gap is smaller than the spread between folds has not beaten the last value; none is even below it.

Almost nothing to fit



What the values do

The model's values stay between -0.50 and 0.32 bp; the changes ran from -16 to 16. Standard deviation 0.120 bp against 4.347. The value had the sign of the change on 48.8% of dates.

The AIC's pick, ARIMA(2, 0, 3), stops on a later fold: did not converge on 755 values.

There is almost nothing to fit, so the model fits almost nothing.

Case study: the question

Context

Does an ARIMA fitted to the past changes get closer to each date's change than carrying the last value forward, once scored walk-forward? A description of how models would have scored on past dates: not a view on rates.

Objective

1. Fit ARIMA to three series' daily changes
2. Score against the last value, day 2's numbers
3. Read each gap against how far the folds differ
4. Check the residuals; say where the models fail

Data: the Treasury's par yields (public domain) and Course 3's invented IG spread.

The three changes

Series	Mean abs change	Std dev	Lag-1 autocorr	Ljung-Box p
10 Yr	3.7480	5.0285	-0.0075	0.0225
2s10s	2.4365	3.3230	-0.0219	0.1461
IG spread (invented)	0.8438	1.1890	-0.0486	0.0024

Series	Best by BIC	Best by AIC	Failed fits
10 Yr	(0, 0)	(2, 3)	8
2s10s	(0, 0)	(0, 2)	5
IG	(0, 0)	(2, 1)	8

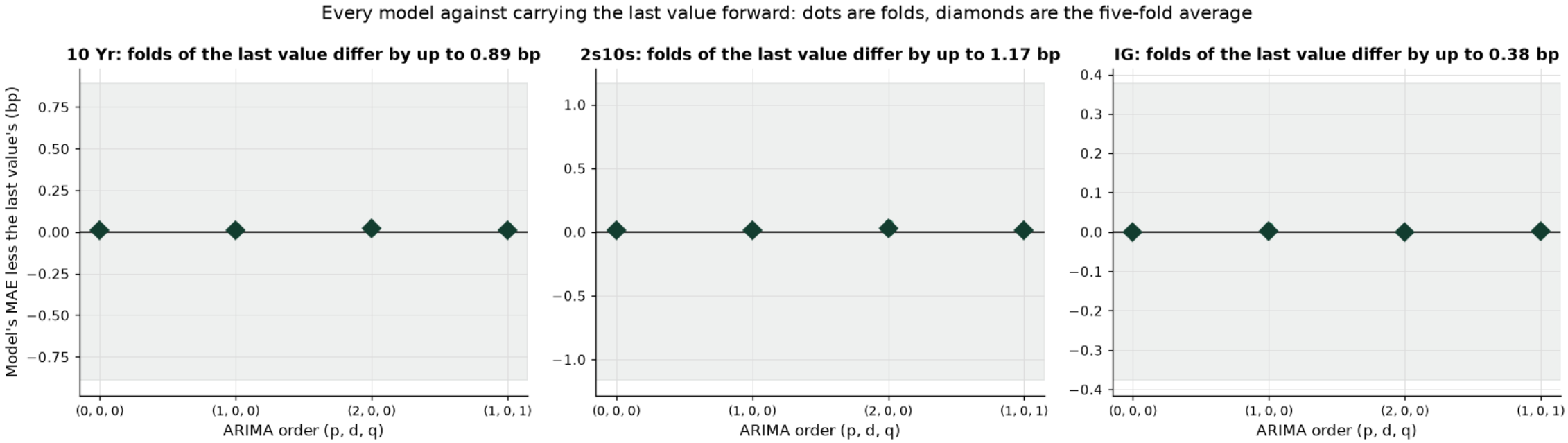
BIC picks the constant on all three; AIC picks longer orders, some of which fail to converge.

Three series, four models

Series	Model	Gap (bp)	MASE	Folds below	Fold range (bp)
10 Yr	(0, 0, 0)	0.0076	1.0024	0	0.8929
10 Yr	(1, 0, 0)	0.0089	1.0028	1	0.8929
10 Yr	(2, 0, 0)	0.0189	1.0059	0	0.8929
10 Yr	(1, 0, 1)	0.0111	1.0035	0	0.8929
2s10s	(0, 0, 0)	0.0134	1.0062	0	1.1667
2s10s	(1, 0, 0)	0.0137	1.0064	0	1.1667
2s10s	(2, 0, 0)	0.0274	1.0127	0	1.1667
2s10s	(1, 0, 1)	0.0155	1.0072	1	1.1667
IG	(0, 0, 0)	0.0003	1.0003	0	0.3778
IG	(1, 0, 0)	0.0024	1.0030	2	0.3778
IG	(2, 0, 0)	0.0001	1.0002	2	0.3778
IG	(1, 0, 1)	0.0010	1.0012	3	0.3778

Every MASE is above 1. Gap: a model's MAE less the last value's, five-fold average. Fold range: how far the last value's own folds differ.

Every gap against its band



Some folds go the other way, and the average is still above zero. Every gap is a small fraction of its band.

Dots are folds; diamonds are five-fold averages; the grey band is how far the last value's own folds differ.

What the models show, and where they fail

What the models show

On the 10 Yr, the 2s10s, and the invented IG spread, ARIMA models of the daily change were no closer than the last value carried forward on any of the five-fold averages: MASE from 1.0002 to 1.0127.

Where they fail

Memory: the AR(1) coefficient is about one standard error from zero

Order: BIC picks the empty model; AIC's picks stop converging

Residuals: heavy tails, and squares with strong memory

Folds: they differ by far more than any gap

A walk-forward error on past dates is not evidence about any later date, and not a view on rates.

Excel side by side

Excel side by side (tested in Excel)

`=SLOPE(D4:D505,D3:D504)`	phi -0.051370
`=INTERCEPT(D4:D505,D3:D504)`	constant 0.069216
`=H2/(1-H1)`	average 0.0658
`=SUMPRODUCT(ABS(D506:D757-($H$2+$H$1*D505:D756)))/ROWS(D506:D757)`	fold 1 MAE 2.7574
`=SUMPRODUCT(ABS(D506:D757))/ROWS(D506:D757)`	last value 2.7619
`=H4/H5`	MASE 0.9984

Excel has no ARIMA fitter: it has the AR(1) as SLOPE and INTERCEPT, and the one-step value as a formula.

Save the day: a file's history

1. Init

The work folder, local settings, the guard

2. Commits

The code, then the 10 Yr's results

3. Grow

Add the other two series to the same file, commit

4. History

git show HEAD~1:file, and git log -- file

```
Day 5: the case study's three series
Day 5: tskit with fit_arima, arima_grid, and walk_forward_arima
```

A results file with a history is a results file you can audit.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which dates fit, which dates score, what may be filled: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the ARIMA is fitted to the levels and scored against the changes: it returns 157.62 bp for a series whose typical change is 0.93 bp.

Code that runs is not the same as code that is right.

What you can do now

- AR, MA, and ARIMA, by hand and with statsmodels
- An order from the AIC and BIC grid, with failed fits counted
- Every setting of ARIMA, and d = 1 on levels is an AR on changes
- Residuals: ACF, Ljung-Box with model_df, Q-Q, squares
- Walk-forward ARIMA against the last value: no model won

Week 2, Day 1 Exponential smoothing